

Luminosity Determination with Restgas Target

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Why to Measure Luminosity

$$N = L \cdot \sigma$$

Number of Events

Time-integrated Luminosity

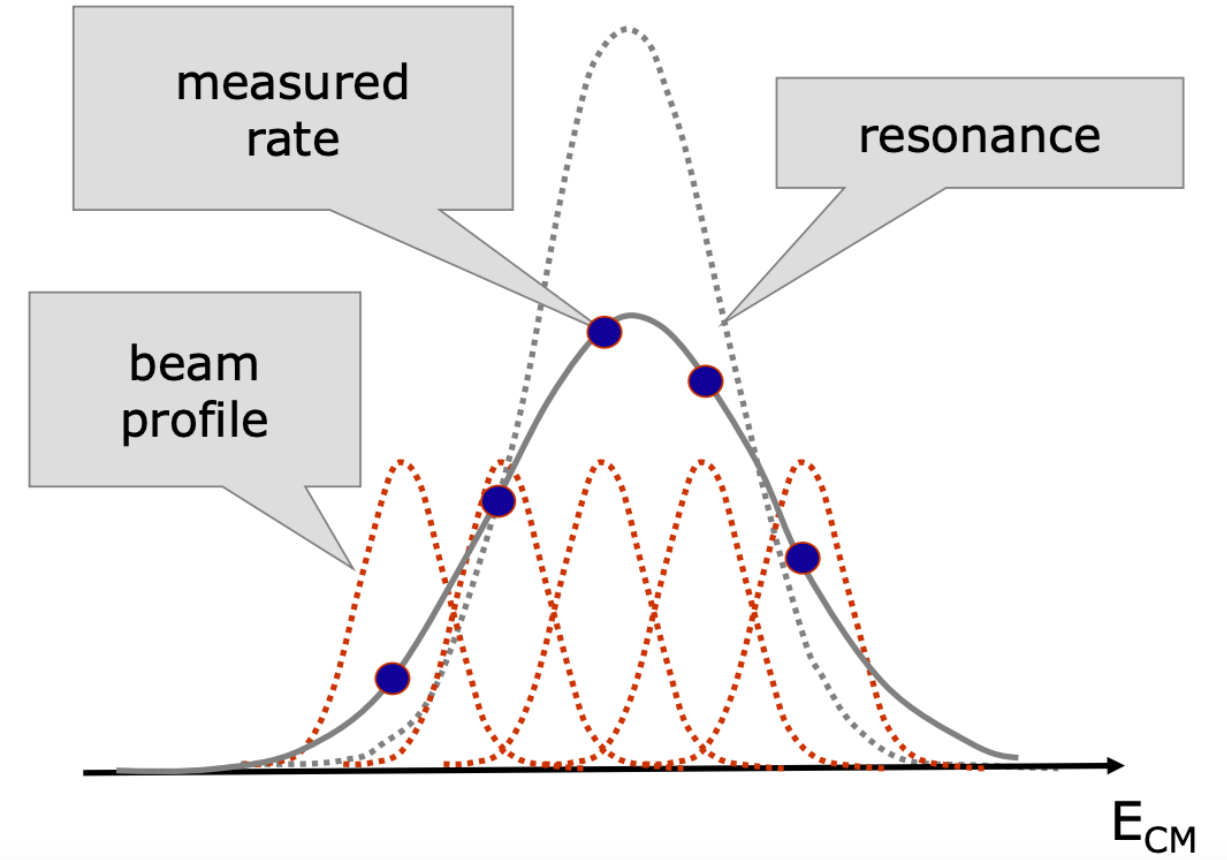
Cross Section

- Absolute cross sections

⇨ Absolute time-integrated luminosity

- Scan experiments

⇨ Relative time-integrated luminosity



How to Measure Luminosity

⇨ $\bar{p}p$ elastic scattering used, only $\bar{p}$ measured

$$\frac{dN(\theta)}{d\theta} = L \cdot \left[\frac{d\sigma(\theta)}{d\theta} \cdot \epsilon(\theta) \right] \otimes Res(\theta)$$

Number of Events

Time-integrated Luminosity

Differential Cross Section

Reconstruction Efficiency

Detector Resolution

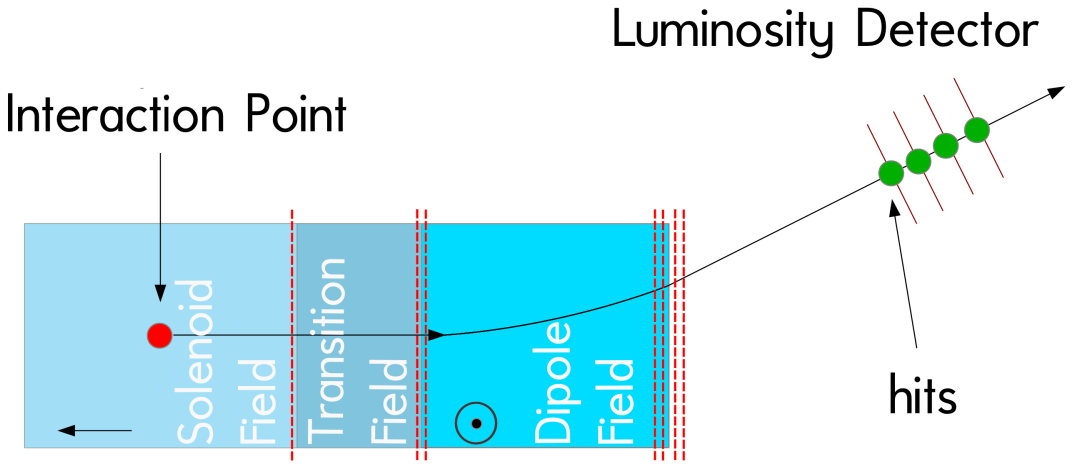
⇨ Determine efficiency and resolution

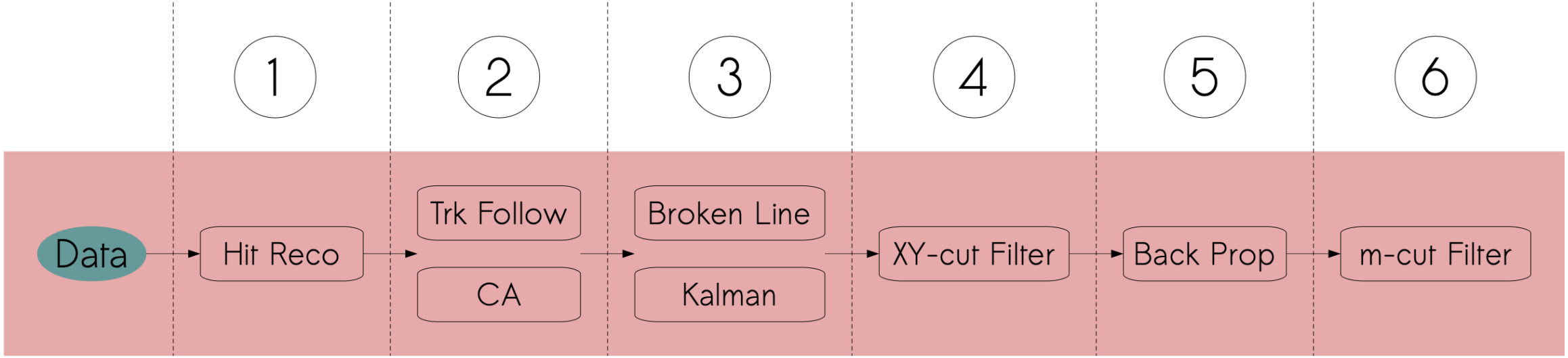
⇨ Determine Luminosity by comparing data and model

Track Reconstruction Software

(PandaRoot/macro/detectors/lmd)

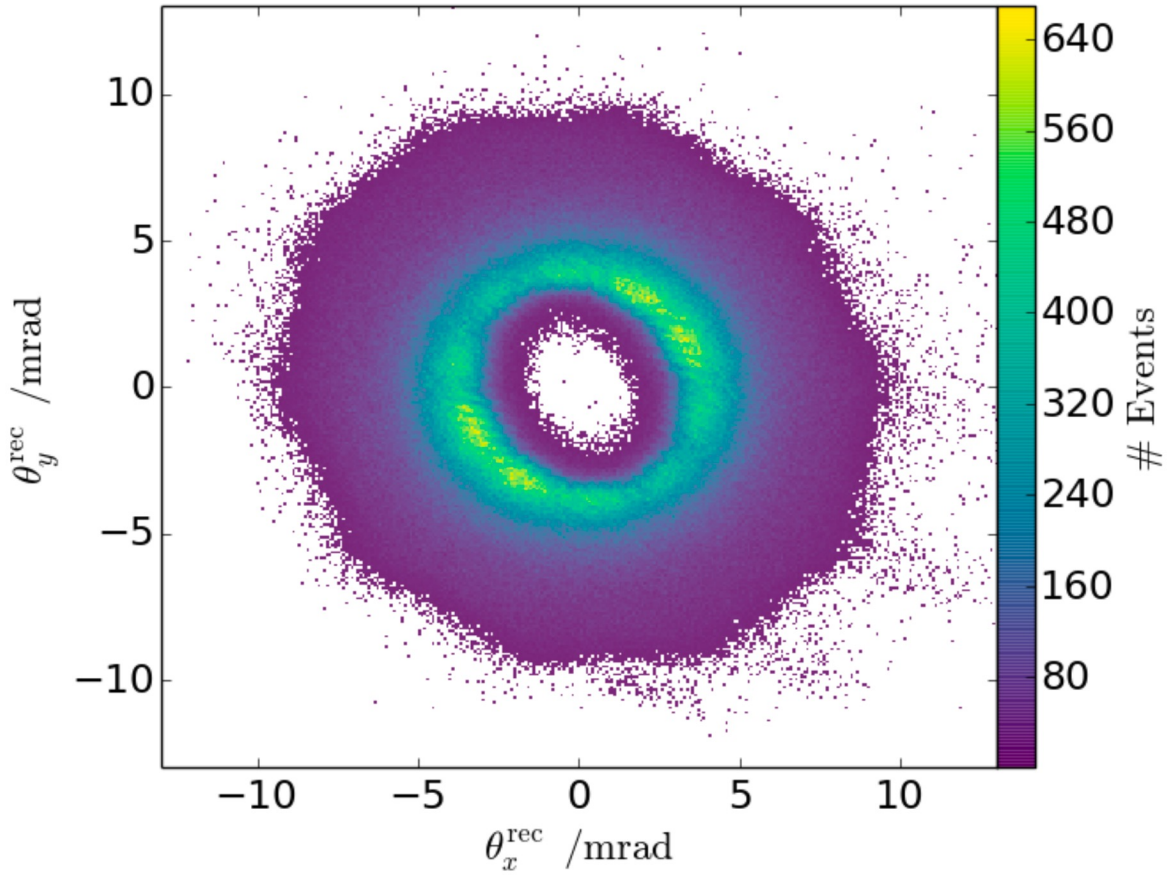
- Sensor data





Track Reconstruction Software

- Final Data → Distribution of tracks at IP

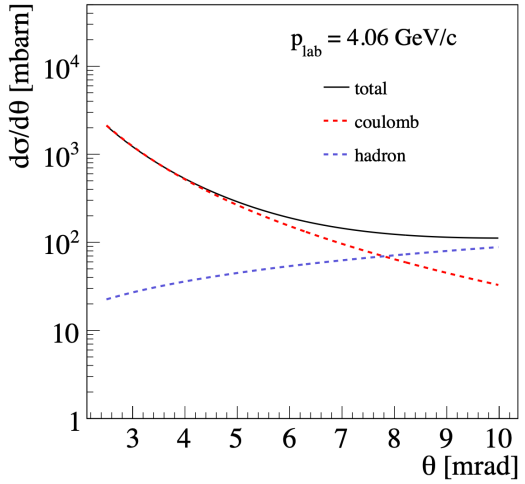


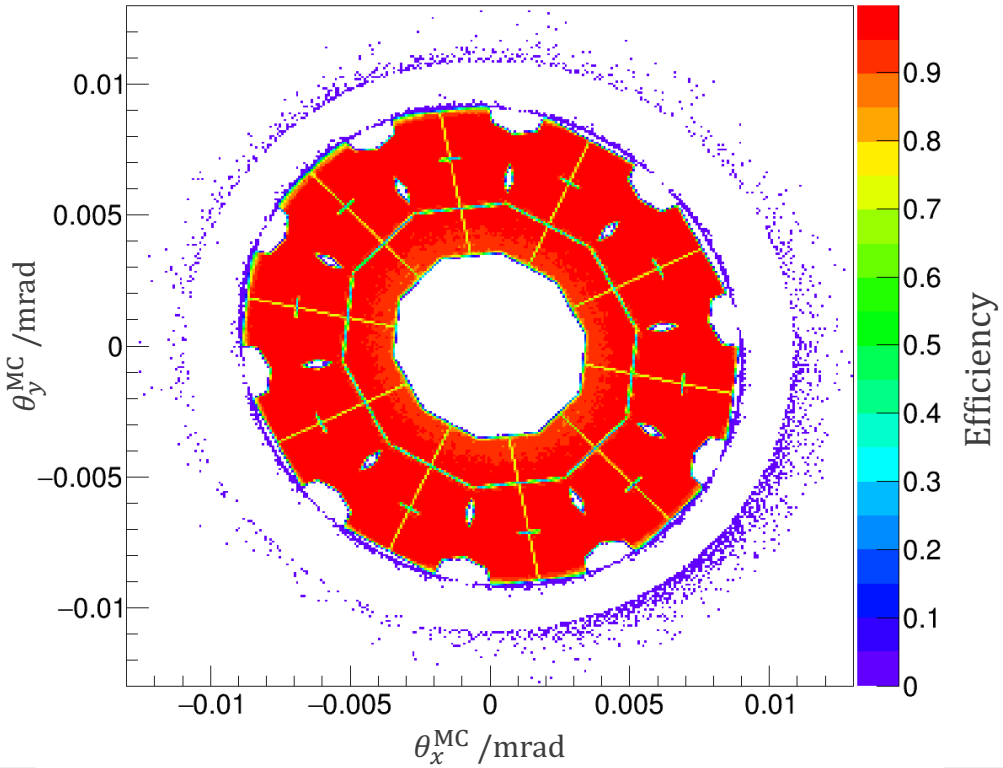
Luminosity Fit Software

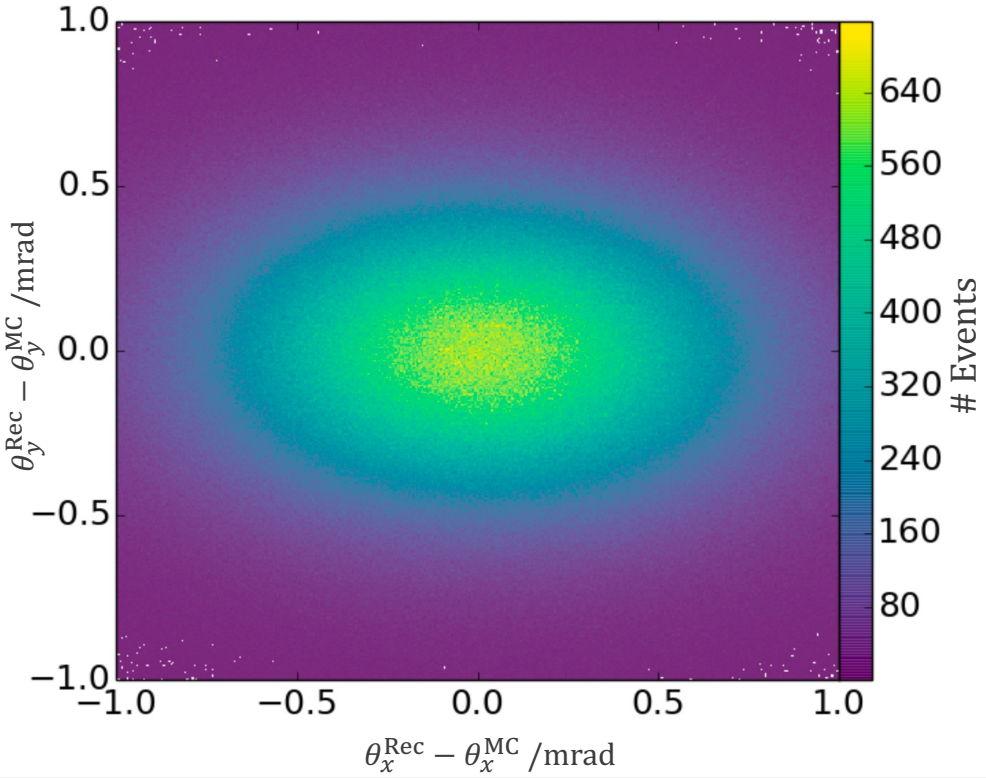
(github.com/LuminosityFit)

- Preparation of the fit model

⇨ Lmd Acceptance & Resolution corrections

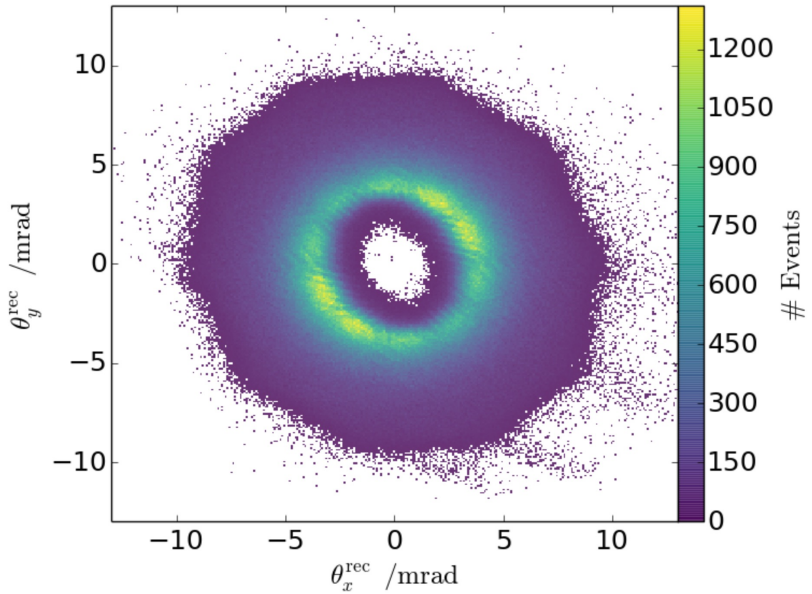






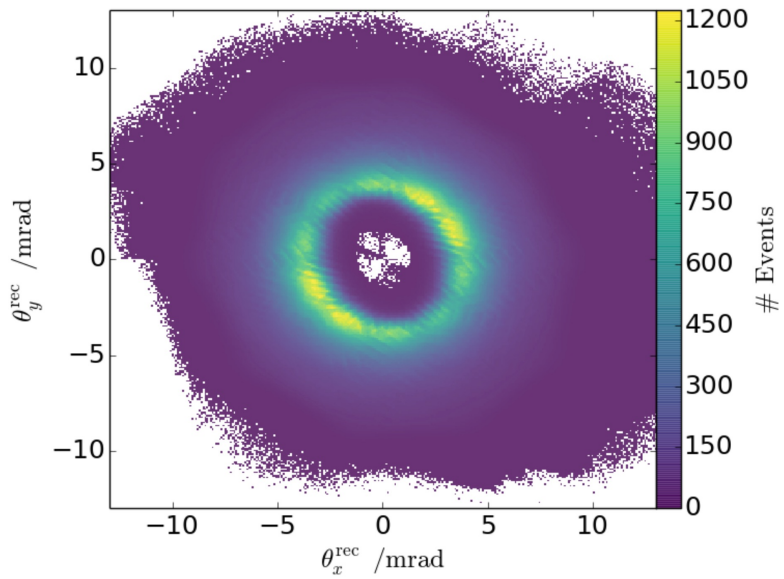
Luminosity Fit Software

- Fit MC model to data distribution



Reconstructed data:

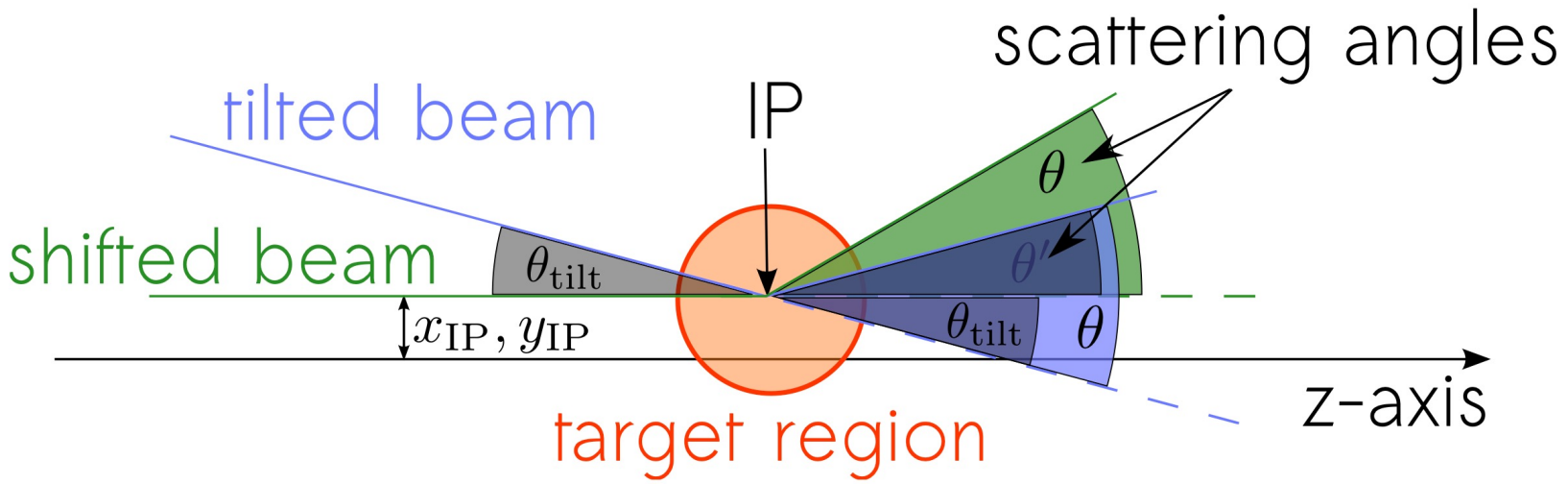
- Back propagated to IP
- Filtering



Fit model:

- Based on elastic cross section
- Corrected for efficiency and resolution

Luminosity Fit Software —— Complications



- determine IP ⇒ acceptance & resolution correction
  - IP position: measured by reconstructed tracks
  - IP distribution: taken from target and accelerator group
- determine beam corrections
  - tilt
  - divergence

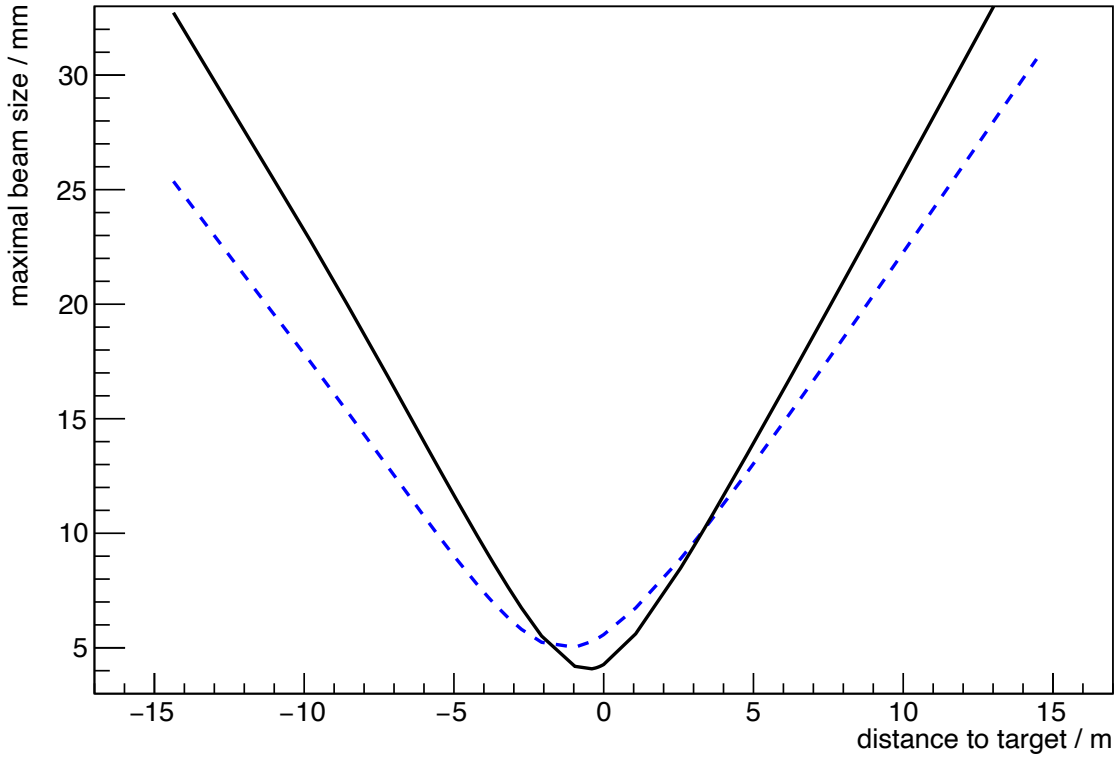
} Determined by LuminosityFit Software

Tilt and Divergence of HESR Beam

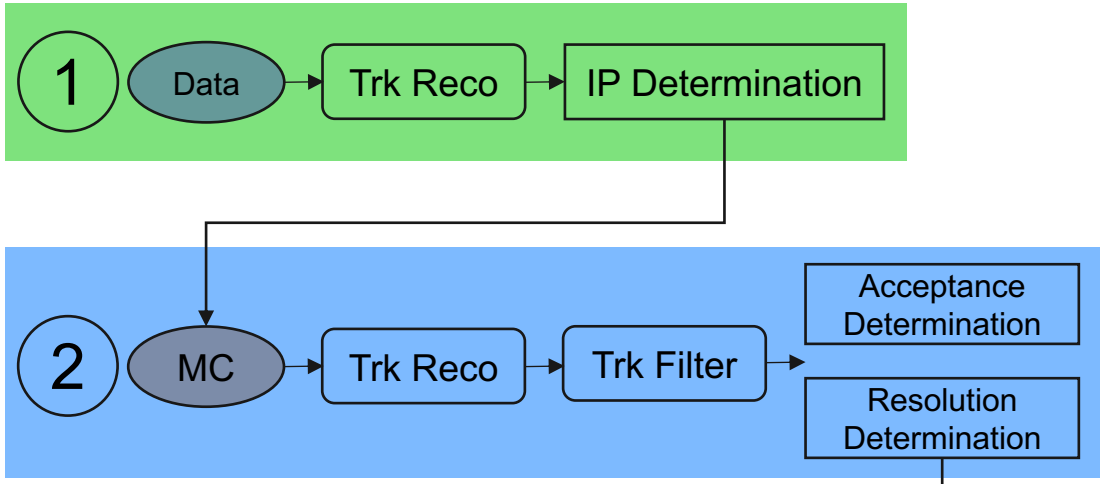
From HESR beam group in COSY

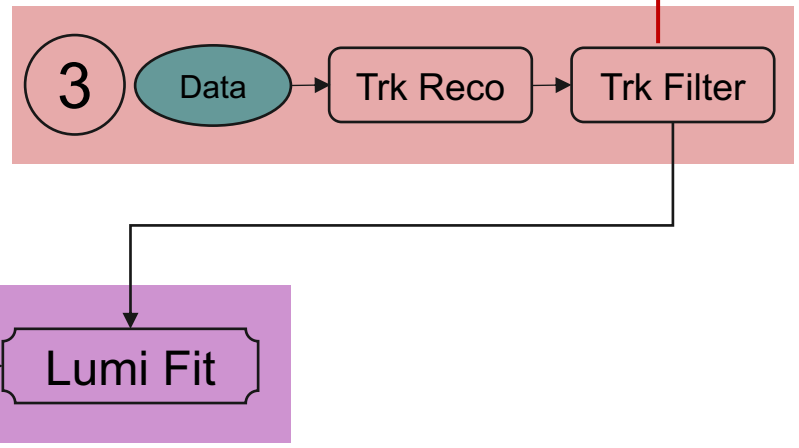
Maximum divergence:
- x direction: 0.9 mm/m
- y direction: 0.6 mm/m

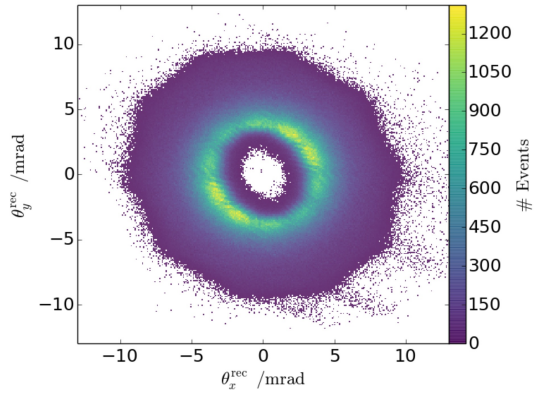
Maximum tilt:
- x direction: 0.8 mm/m
- y direction: 0.3 mm/m

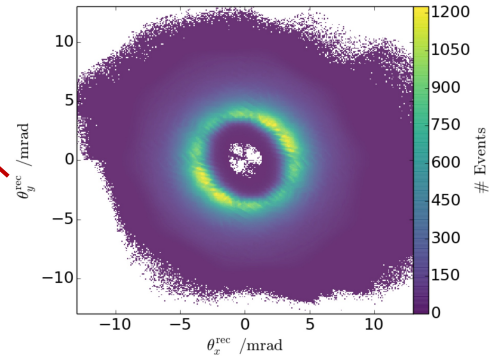


Luminosity Fit Process







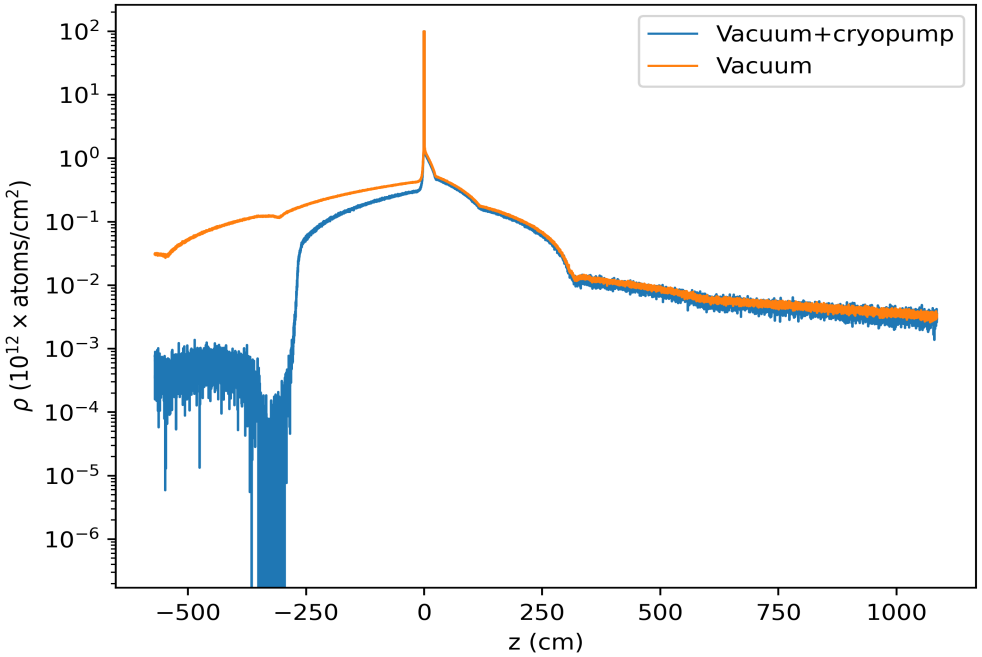


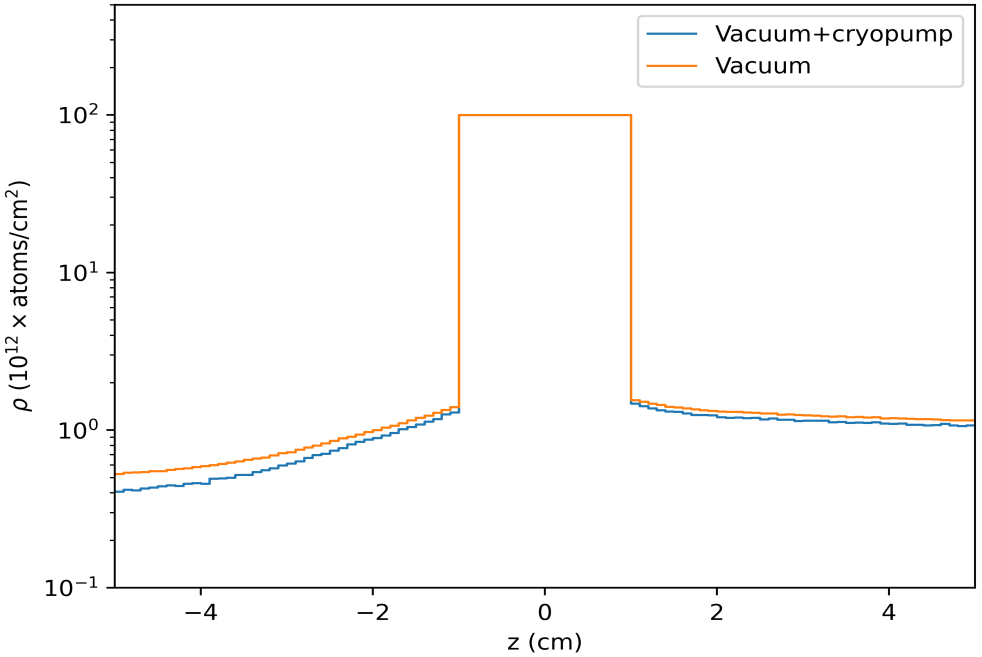
Output:

- Time- integrated Luminosity
- Tilt and divergence of the beam

Restgas Distribution

From target group in Muenster



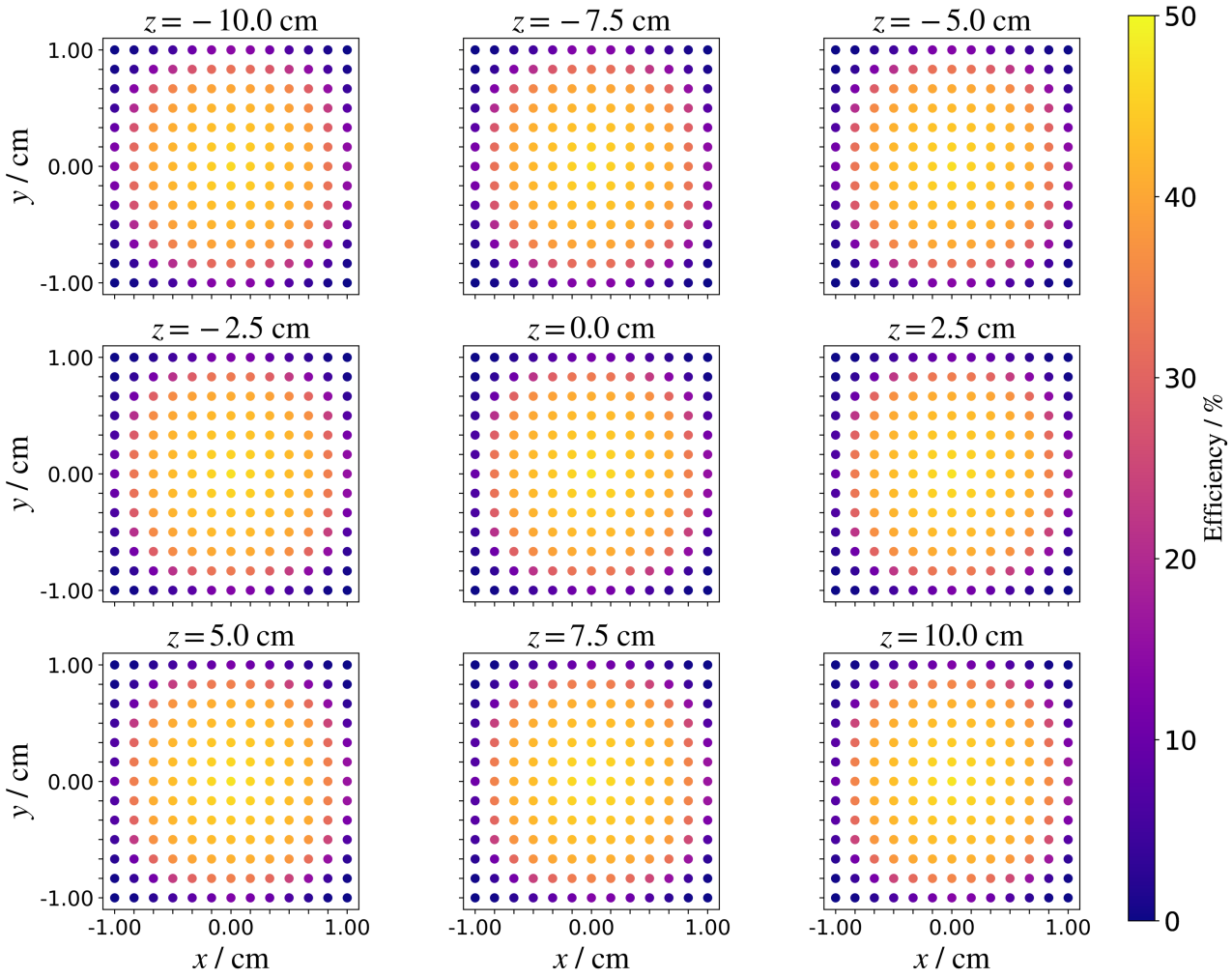


Luminosity detector

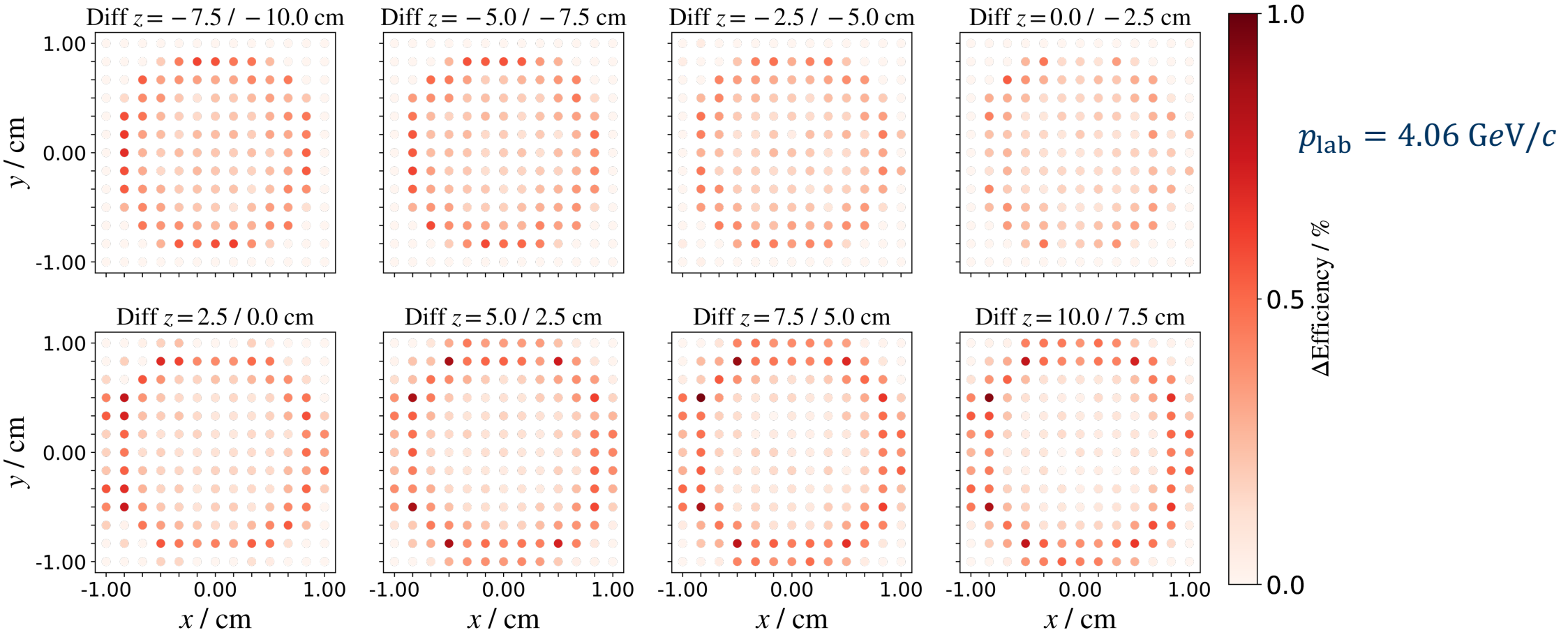
- Track reconstruction not sensitive to z position of the origin of the tracks

Efficiency Dependent on IP

$p_{\mathrm{lab}} = 4.06\ \mathrm{GeV}/c$



Efficiency Difference Between Two z Positions

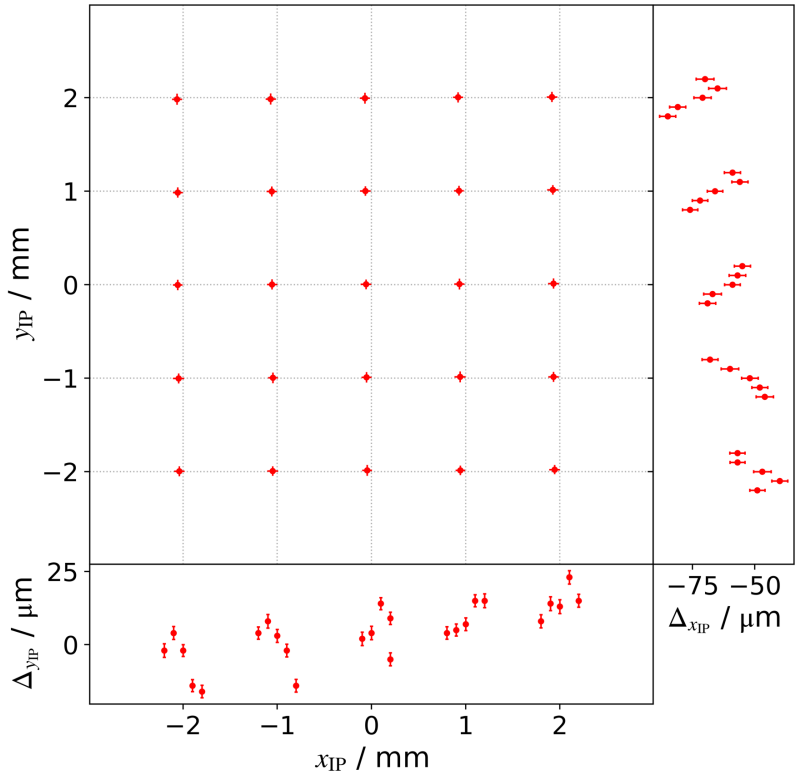


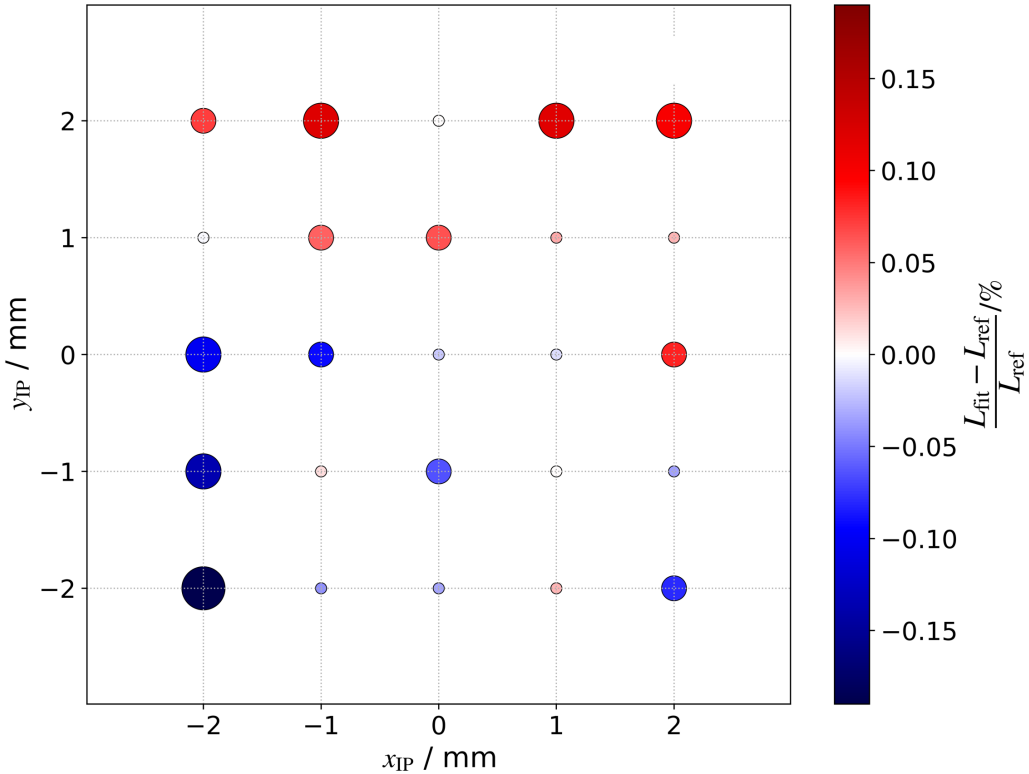
⇨ Efficiency increase with z

Check with the Restgas Target

Scenarios with IP shift only

- Determination of IP shift $(p_{\mathrm{lab}} = 4.06\ \mathrm{GeV}/c)$





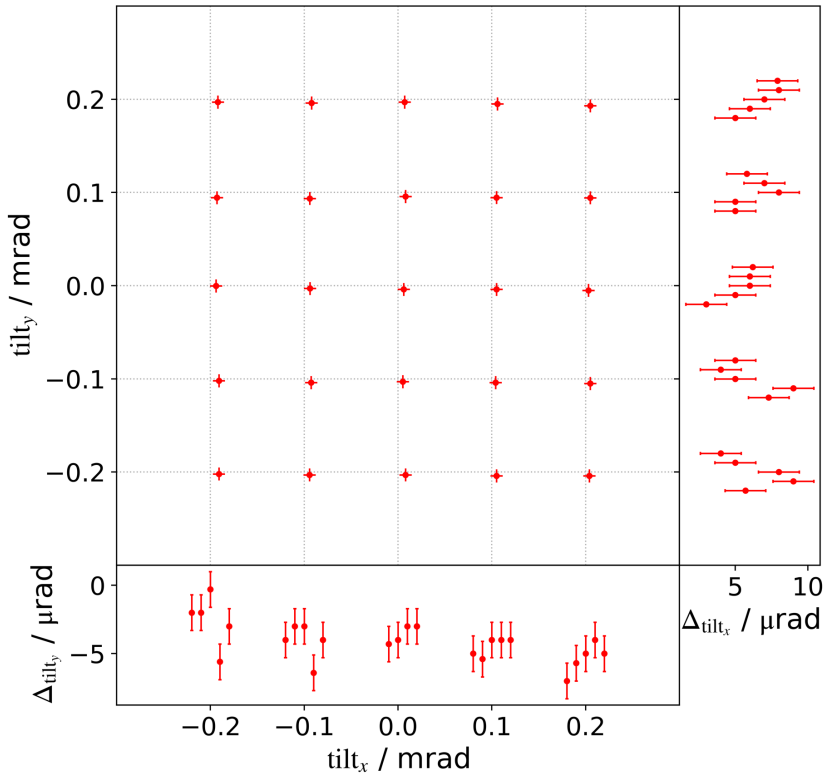
⇨ Maximum deviation after correction:

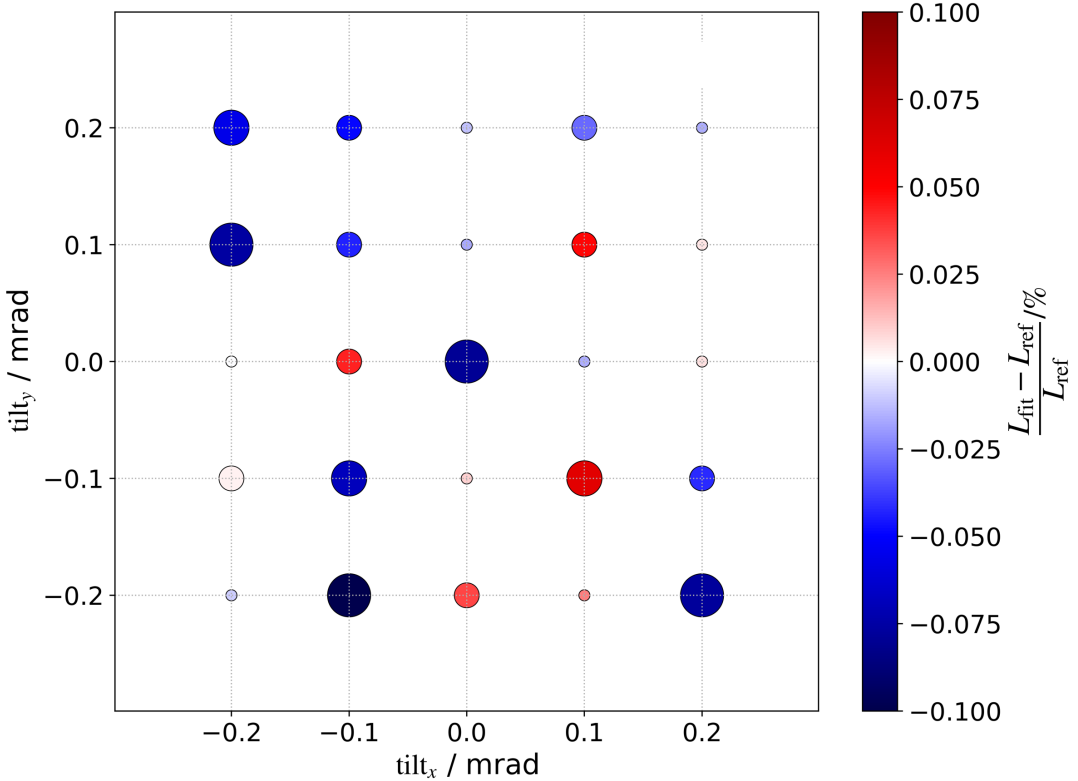
$|\Delta x| \leq 85\ \mu\mathrm{m}$ $|\Delta y| \leq 25\ \mu\mathrm{m}$ $|\Delta L| < 0.20\ \%$

Check with the Restgas Target

Scenarios with beam tilt only

- Determination of beam tilt $(p_{\mathrm{lab}} = 4.06\ \mathrm{GeV}/c)$





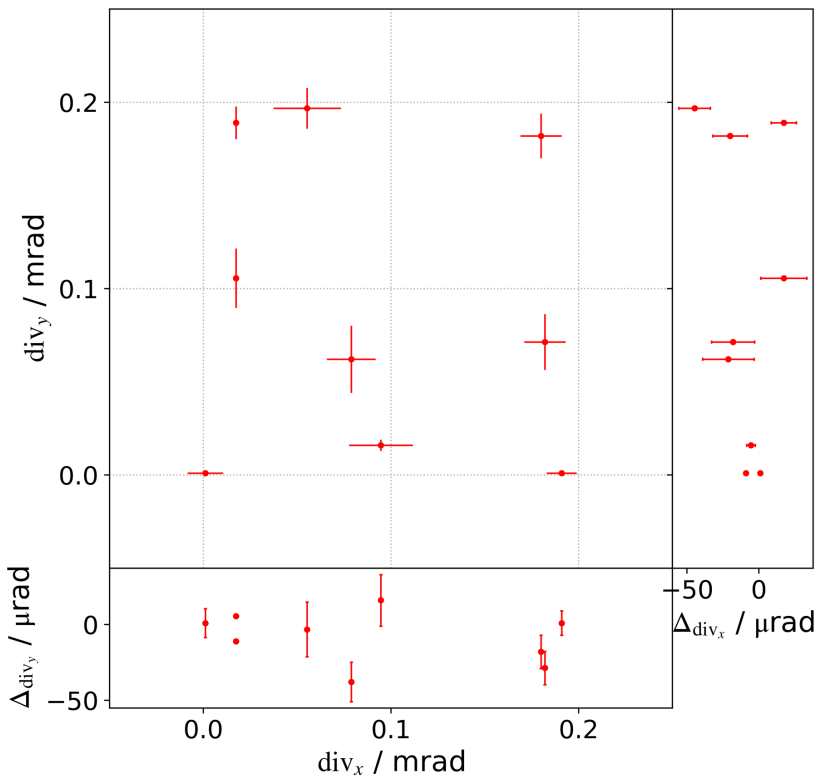
⇨ Maximum deviation after taking into account:

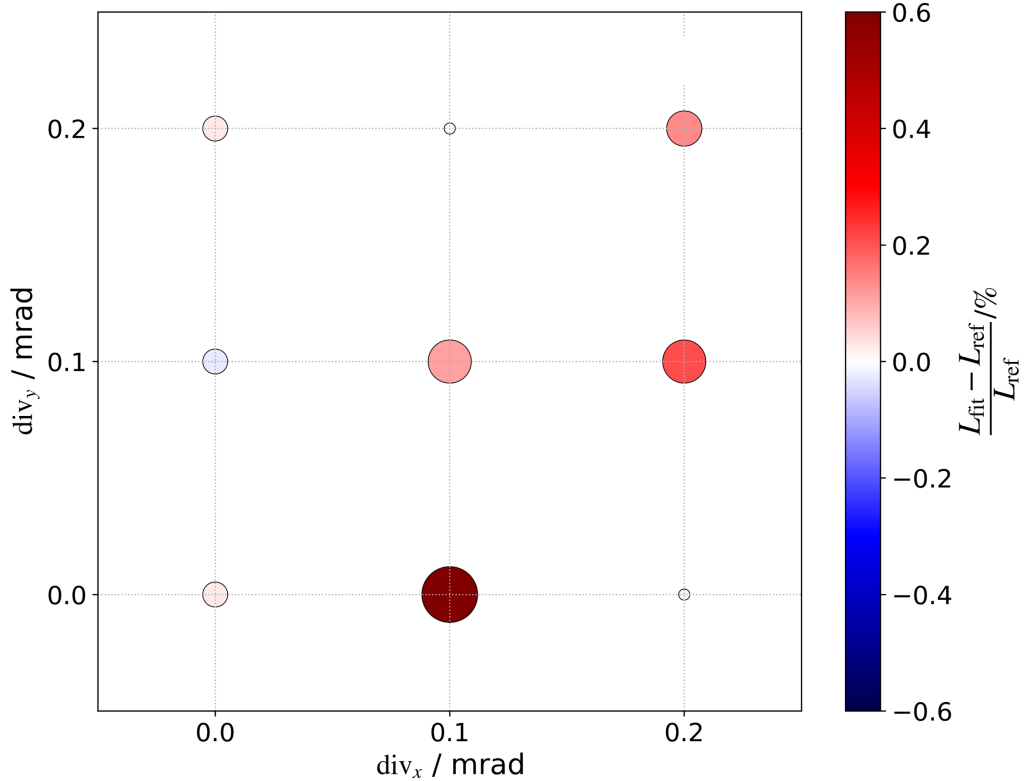
$|\Delta\theta_x| \leq 7\ \mu\mathrm{rad}$ $|\Delta\theta_y| \leq 9\ \mu\mathrm{rad}$ $|\Delta L| < 0.11\ \%$

Check with the Restgas Target

Scenarios with beam divergence only

- Determination of beam divergence $(p_{\mathrm{lab}} = 4.06\ \mathrm{GeV}/c)$





⇨ Maximum deviation after taking into account :

$|\Delta \mathrm{div}_x| \leq 40\ \mu\mathrm{rad}$ $|\Delta \mathrm{div}_y| \leq 45\ \mu\mathrm{rad}$ $|\Delta L| < 0.70\ \%$

Check of Fit Procedure

Fit steps:

- Include IP position
- Fit Parameters:
  - Beam tilt
  - Beam divergence
  - Luminosity

Results:

- Tilt results quite good
- Luminosity not too bad

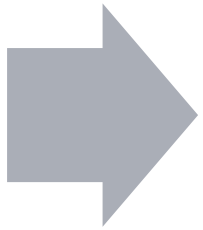
- Fit with divergence fixed
- $p_{\mathrm{lab}} = 4.06\ \mathrm{GeV}$

tilt_x	tilt_y	div_x	div_y	$\Delta L/L$
	(μrad)			(%)
100	100	100	100	0.30 ± 0.07
105	96.3	0	0	
100	100	100	100	-0.05 ± 0.07
105	95.9	100	100	
100	100	100	100	-1.05 ± 0.07
104	94.5	200	200	

Analysis Procedure

Fit with a certain start divergence

$\Longrightarrow$ Determination of tilt



Fit with fixed tilt

$\Longrightarrow$ Determination of divergence

Luminosity Fit Results with Cutting Around IP

- $p_{\mathrm{lab}} = 4.06\ \mathrm{GeV}/c$
- IP cut around reconstructed IP:

  1. $x, y < 0.5\ \mathrm{cm}, z < 1\ \mathrm{cm}$ 2. $x, y < 0.1\ \mathrm{cm}, z < 0.3\ \mathrm{cm}$

- Corrected generated luminosity by $\mathrm{ratio} = \frac{\text{Number of MC events after IP cut}}{\text{Number of generated MC events}}$

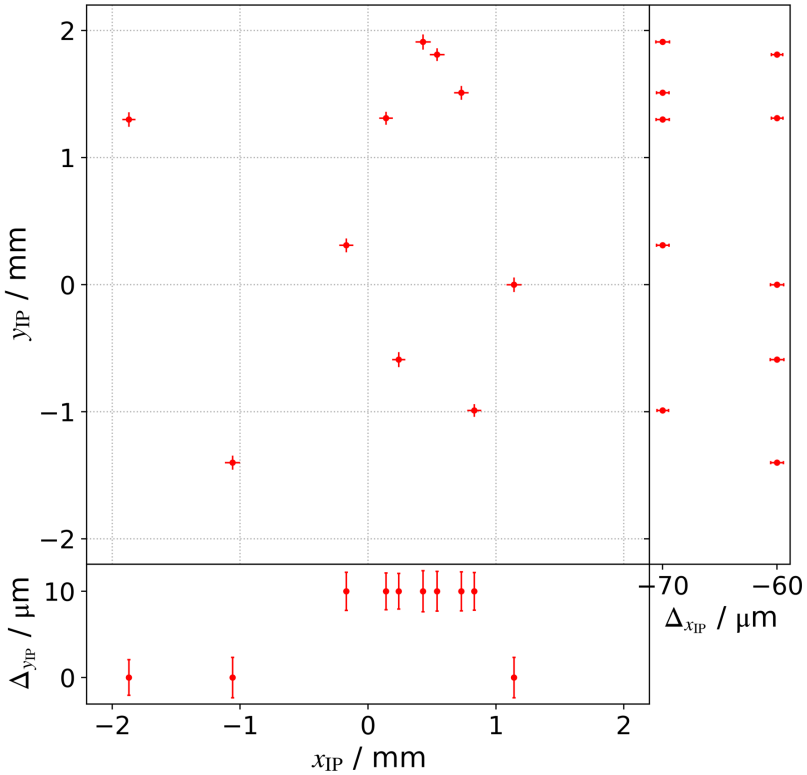
Generated Luminosity	Corrected Generated Luminosity	Measured Luminosity	Corrected $\Delta L/L$ (%)	$\Delta L/L$ without correction (%)
3.568×10^6	2.225×10^6	2.226×10^6	-0.001 ± 0.003	-37.61 ± 0.05
3.587×10^6	4.853×10^5	4.844×10^5	-0.006 ± 0.18	-86.49 ± 0.02

⇨ Luminosity quite good with different cuts

Comparison w/o Restgas (Realistic Scenarios $p = 4.06\ GeV/c$)

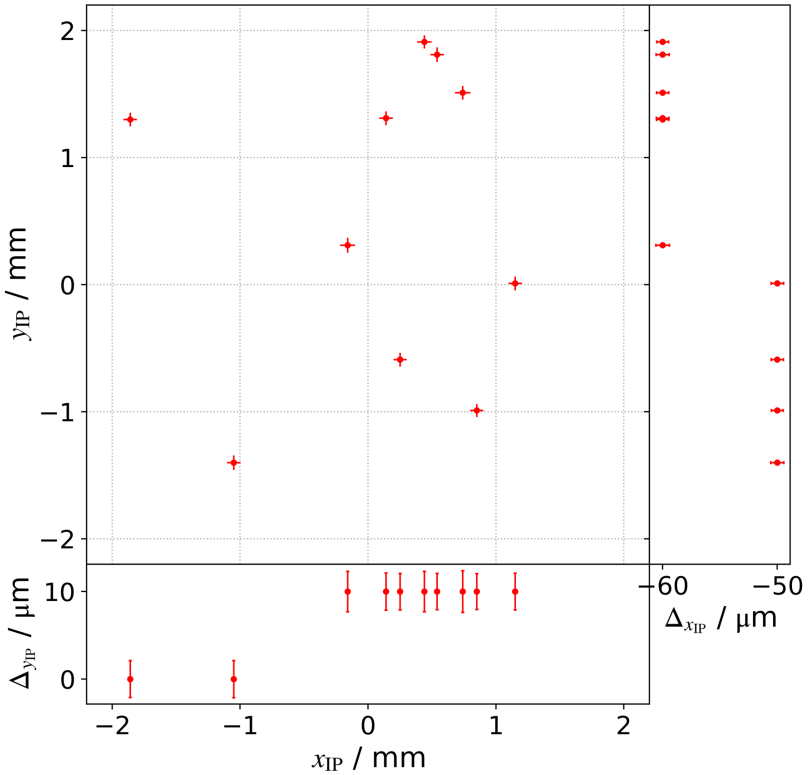
IP shift

With Restgas



⇨ Maximum deviation:
$|\Delta x| \leq 70\ \mu m \quad |\Delta y| \leq 10\ \mu m$

Without Restgas

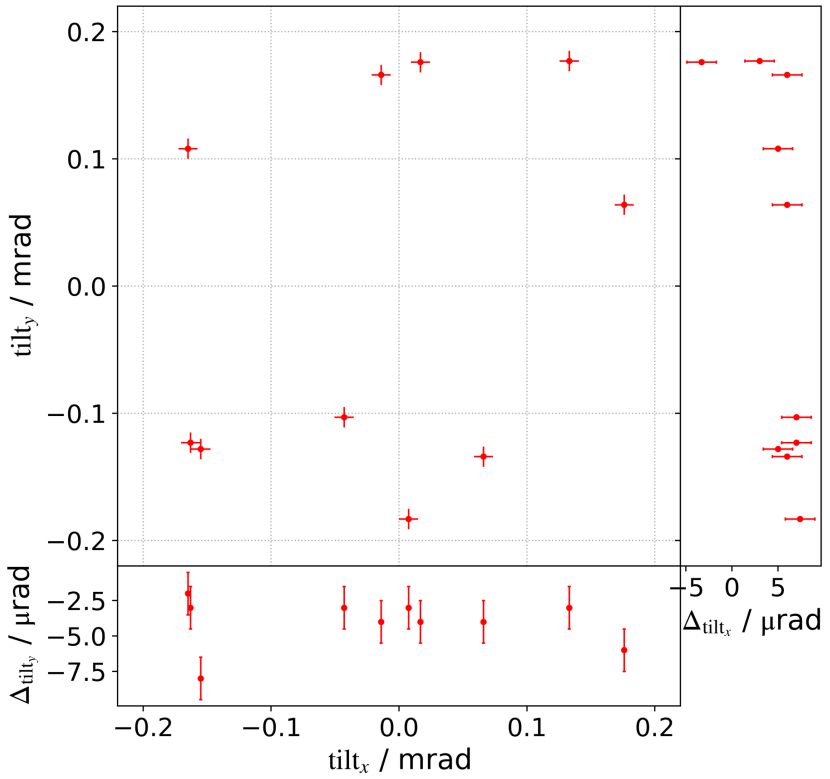


⇨ Maximum deviation:
$|\Delta x| \leq 60\ \mu m \quad |\Delta y| \leq 10\ \mu m$

Comparison w/o Restgas (Realistic Scenarios $p = 4.06\ \mathrm{GeV}/c$)

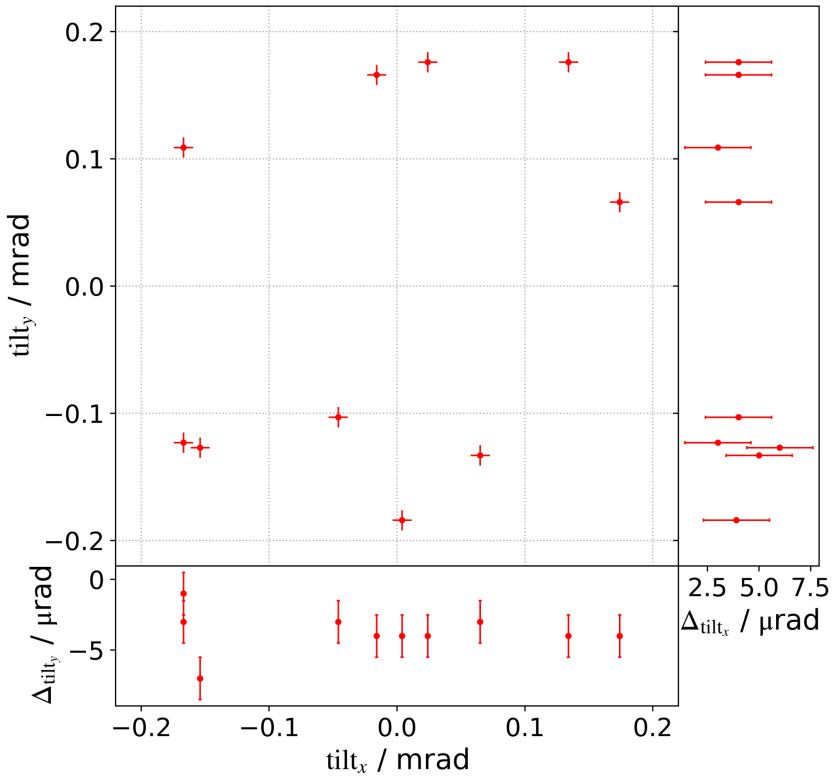
Beam tilt

With Restgas



⇨ Maximum deviation:
$|\Delta\theta_x| \leq 6\ \mu\mathrm{rad}$ $|\Delta\theta_y| \leq 7\ \mu\mathrm{rad}$

Without Restgas

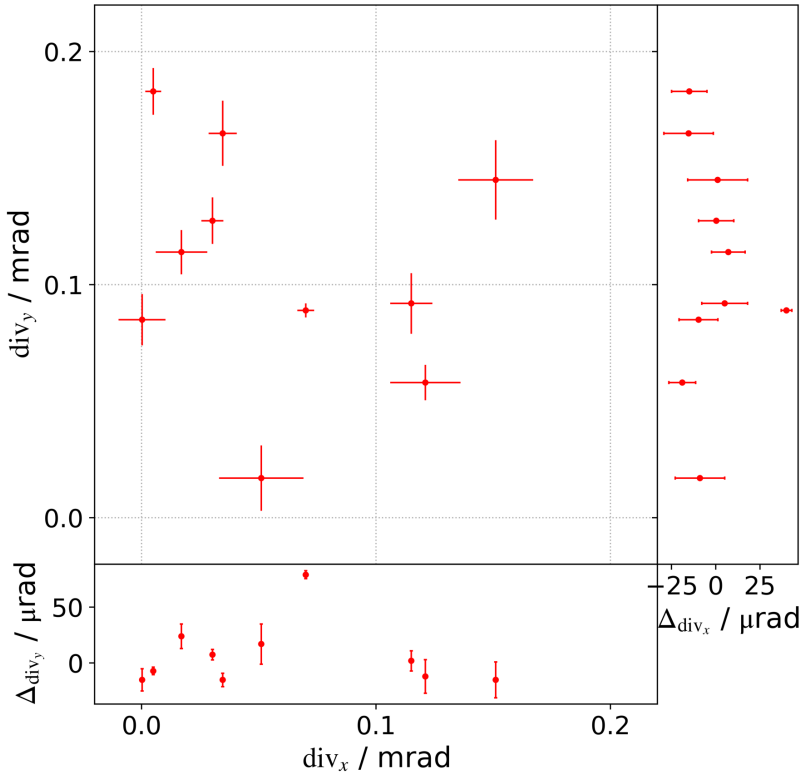


⇨ Maximum deviation:
$|\Delta\theta_x| \leq 8\ \mu\mathrm{rad}$ $|\Delta\theta_y| \leq 8\ \mu\mathrm{rad}$

Comparison w/o Restgas (Realistic Scenarios $p = 4.06\ GeV/c$)

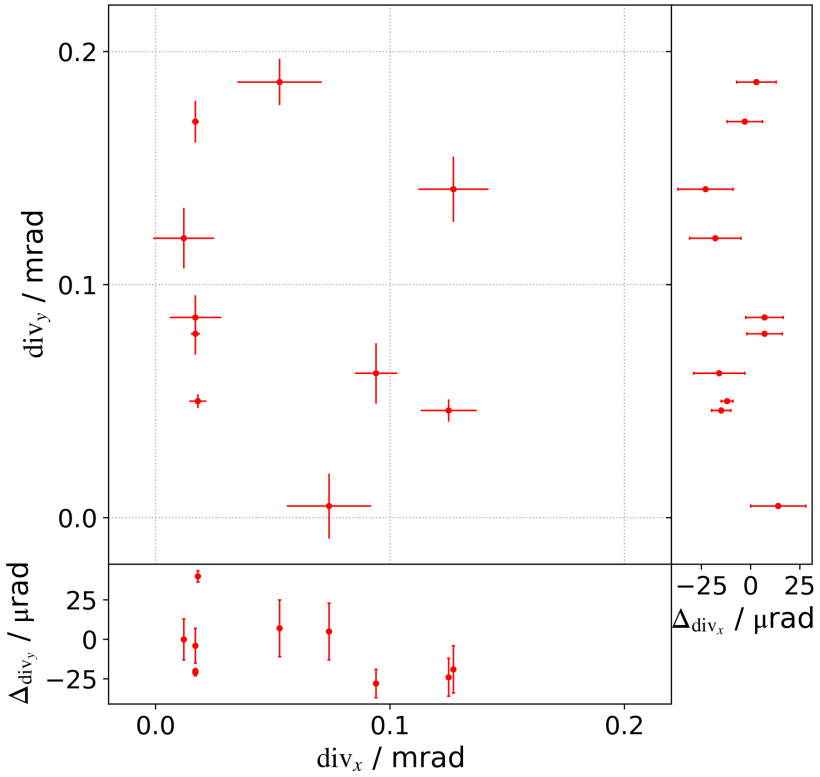
Beam divergence

With Restgas



⇨ Maximum deviation:
$|\Delta\mathrm{div}_x| \leq 40\ \mu\mathrm{rad}$ $|\Delta\mathrm{div}_y| \leq 80\ \mu\mathrm{rad}$

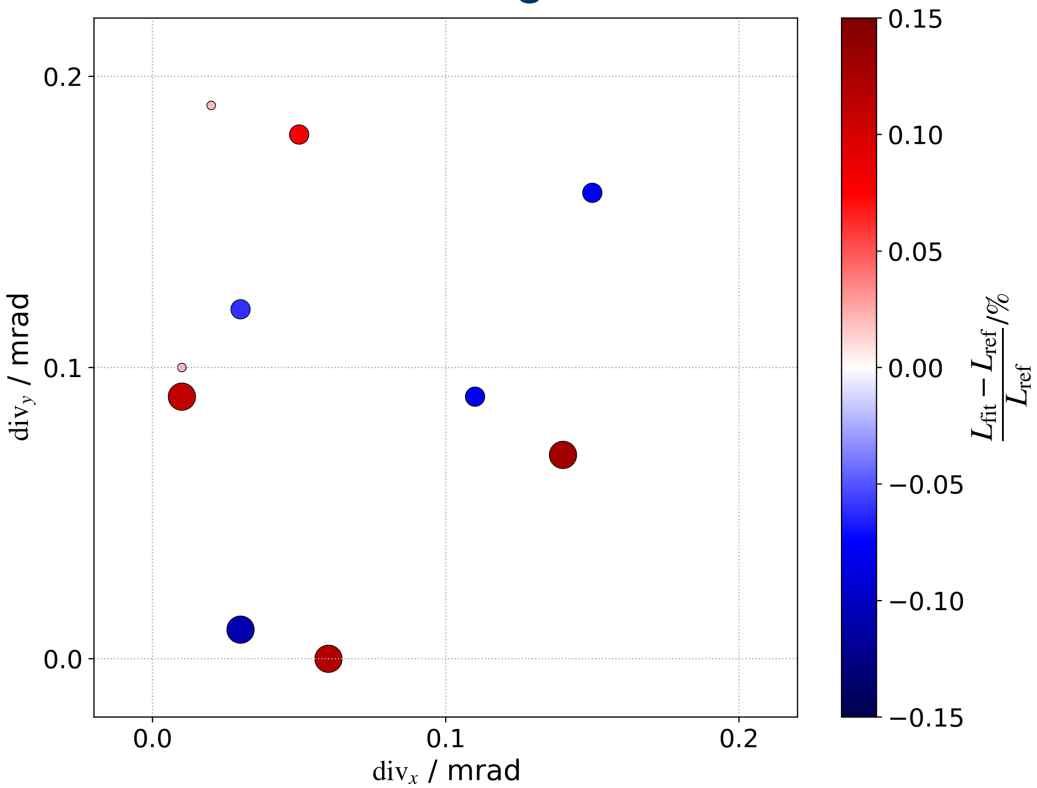
Without Restgas



⇨ Maximum deviation:
$|\Delta\mathrm{div}_x| \leq 25\ \mu\mathrm{rad}$ $|\Delta\mathrm{div}_y| \leq 40\ \mu\mathrm{rad}$

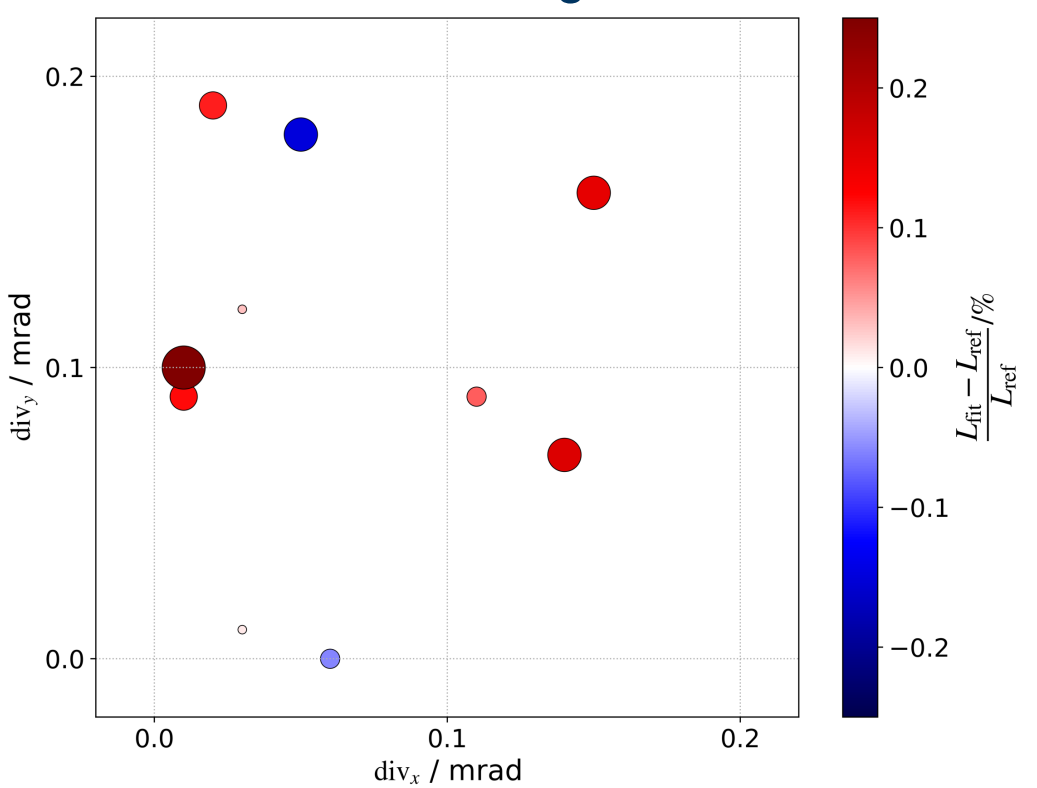
Comparison w/o Restgas (Realistic Scenarios $p = 4.06\ GeV/c$)

With Restgas



⇨ Maximum deviation:
$|\Delta L| \leq 0.15\%$

Without Restgas



⇨ Maximum deviation:
$|\Delta L| \leq 0.25\%$

Summary and Outlook

- Study with Restgas:

	Without Rest Gas	With Rest Gas
$\lvert\Delta x\rvert$ / $\lvert\Delta y\rvert$	< 60 µm / ~ 10 µm	< 70 µm / ~ 10 µm
$\lvert\Delta\theta\rvert$	< 10 µrad	< 10 µrad
$\lvert\Delta\mathrm{div}\rvert$	< 50 µrad	< 50 µrad
$\lvert\Delta L\rvert$	$< 0.25\%$	$< 0.15\%$

- Cutting around IP is applied in Luminosity Fit

⇨ Relative deviation of Luminosity: $< 0.5\%$ for p_{lab} $1.5\mathrm{GeV}/c - 15\mathrm{GeV}/c$

⇨ Measurement of the restgas distribution by using elastic scattering (Determination of the fraction) for further studies



Thanks For Your Attention!

Backups

Luminosity Determination in Realistic Scenarios

p_{lab}	x	y	x, y_{cut}	z_{cut}	tilt_x	tilt_y	div_x	div_y	$\Delta L/L$
(GeV/c)		(mm)				(μrad)			(%)
1.50	-2.0	1.2	6	60	-150	-40	120	50	0.11 ± 0.11
	-2.12	1.23			-144	-48.8	84.9	17.3	
	-1.6	0.6	4	40	-180	-110	50	130	-0.18 ± 0.22
	-1.69	0.62			-174	-116	74.2	137	
	-0.5	-0.3	4	10	-80	0	80	100	0.16 ± 0.24
	-0.62	-0.28			-73.3	-10.3	59.1	78.6	
	1.3	0.6	12	20	-100	190	200	0	-0.09 ± 0.11
	1.16	0.63			-90.2	178	199	3	
	0.3	-1.8	5	30	-60	-130	120	20	0.13 ± 0.08
	0.18	-1.77			-53.1	-140	139	17.7	

Luminosity Determination in Realistic Scenarios

p_{lab}	x	y	x, y_{cut}	z_{cut}	tilt_x	tilt_y	div_x	div_y	$\Delta L/L$
(GeV/c)		(mm)				(μrad)			(%)
1.50	-0.5	-1.6	9	40	150	40	30	130	0.20 ± 0.17
	-0.62	-1.57			158	30	5.2	146	
	1.0	2.0	7	30	170	-90	160	60	-0.10 ± 0.11
	0.87	2.04			-171	-104	162	12.3	
	2.0	1.1	10	50	30	120	140	40	0.007 ± 0.15
	1.86	1.13			39.3	108	107	10.8	
	-1.5	-0.3	4	50	-110	160	40	170	-0.05 ± 0.11
	-1.62	-0.28			-104	155	25.6	154	
	1.4	-0.2	7	10	-10	-130	100	100	-0.54 ± 0.24
	1.27	-0.16			-2.0	-141	147	146	

Luminosity Determination in Realistic Scenarios

p_{lab}	x	y	x, y_{cut}	z_{cut}	tilt_x	tilt_y	div_x	div_y	$\Delta L/L$
(GeV/c)		(mm)				(μrad)			(%)
4.06	-1.3	0.2	10	40	-170	60	80	160	0.58 ± 0.08
	-1.37	0.20			-165	59.6	93.2	144	
	1.0	-1.4	5	60	-120	70	50	140	-0.53 ± 0.64
	0.94	-1.40			-113	65.1	62.1	163	
	-1.1	-1.3	12	10	-30	140	190	20	0.52 ± 0.16
	-1.16	-1.30			-24.5	135	180	33.8	
	1.2	0.1	12	30	150	-10	10	120	0.21 ± 0.08
	1.14	0.11			155	-13.3	34.6	111	
	0.5	0.6	9	30	110	130	90	200	0.16 ± 0.10
	0.44	0.61			116	127	94.9	176	

Luminosity Determination in Realistic Scenarios

p_{lab}	x	y	x, y_{cut}	z_{cut}	tilt_x	tilt_y	div_x	div_y	$\Delta L/L$
(GeV/c)		(mm)				(μrad)			(%)
4.06	0.9	−1.6	12	10	−150	50	60	120	0.25 ± 0.13
	0.84	−1.60			−142	46.2	17.3	169	
	−0.6	−1.4	8	60	−100	110	110	190	0.21 ± 0.11
	−0.66	−1.40			−93.9	104	42.4	173	
	−0.7	−0.1	3	60	30	80	90	100	-0.22 ± 0.16
	−0.76	−0.10			34.5	77.8	119	113	
	0.4	−1.6	12	60	50	−30	70	10	-0.07 ± 0.11
	0.34	−1.60			56.1	−33.5	79.2	4.5	
	0.2	−1.5	3	30	80	130	90	70	0.02 ± 0.11
	0.14	−1.45			80.6	126	79.9	26.1	

Luminosity Determination in Realistic Scenarios

p_{lab}	x	y	x, y_{cut}	z_{cut}	tilt_x	tilt_y	div_x	div_y	$\Delta L/L$
(GeV/c)		(mm)				(μrad)			(%)
8.90	1.3	−1.9	12	60	−120	−120	90	90	-0.04 ± 0.17
	1.23	−1.91			−111	−118	52.0	52.0	
	0.9	0.7	12	20	−160	160	100	90	0.02 ± 0.09
	0.83	0.71			−146	156	97.1	27.2	
	1.2	0.8	6	10	−30	−100	100	110	0.03 ± 0.10
	1.13	0.80			−20.3	−109	95.7	92.9	
	1.8	−0.2	9	60	−20	80	180	140	0.08 ± 0.12
	1.73	−0.20			−6.3	81.6	157	135	
	−0.5	0.5	10	60	−40	70	10	150	0.07 ± 0.09
	−0.58	0.51			−34.1	74.5	54.4	86.8	

Luminosity Determination in Realistic Scenarios

p_{lab}	x	y	x, y_{cut}	z_{cut}	tilt_x	tilt_y	div_x	div_y	$\Delta L/L$
(GeV/c)		(mm)				(μrad)			(%)
8.90	2.0	0.4	3	10	−160	−20	20	160	0.06 ± 0.14
	1.82	0.40			−138	−18.2	8.9	174	
	−0.5	0.7	9	30	20	−90	150	40	0.06 ± 0.09
	−0.58	0.70			26.8	−91.7	173	34.6	
	0.8	0.2	4	50	−120	−40	10	100	0.07 ± 0.11
	0.73	0.20			−113	−41.8	12.2	113	
	1.6	0.7	12	10	−160	−170	50	20	-0.004 ± 0.28
	1.53	0.70			−148	−168	49.5	9.6	
	−0.1	1.8	3	20	10	70	150	140	-0.03 ± 0.12
	−0.17	1.72			8.3	58.2	121	106	

Luminosity Determination in Realistic Scenarios

p_{lab}	x	y	x, y_{cut}	z_{cut}	tilt_x	tilt_y	div_x	div_y	$\Delta L/L$
(GeV/c)		(mm)				(μrad)			(%)
15.0	1.8	1.0	10	20	50	0	80	80	0.09 ± 0.09
	1.70	1.00			78.6	23.3	67.1	73.6	
	0.8	0.1	10	30	60	20	60	10	0.01 ± 0.09
	0.70	0.10			59.5	27.8	93.1	21.8	
	−0.6	1.4	9	30	−140	−150	110	200	0.09 ± 0.09
	−0.70	1.40			−128	−135	181	209	
	1.2	−0.2	5	60	100	190	80	130	0.04 ± 0.09
	1.10	−0.20			122	207	18.4	173	
	−1.6	−0.1	12	60	150	140	190	90	-0.05 ± 0.09
	−1.70	−0.10			147	156	173	76.5	

Luminosity Determination in Realistic Scenarios

p_{lab}	x	y	x, y_{cut}	z_{cut}	tilt_x	tilt_y	div_x	div_y	$\Delta L/L$
(GeV/c)		(mm)				(μrad)			(%)
15.0	-1.7	-0.2	4	10	-130	-180	140	80	-0.09 ± 0.09
	-1.79	-0.20			-146	-178	132	90.4	
	0.5	-0.6	10	20	-80	120	170	180	0.15 ± 0.09
	0.40	-0.60			-75.4	135	152	194	
	1.5	-0.5	3	50	-120	-40	0	70	0.07 ± 0.09
	1.36	-0.50			-123	-49.4	5.1	100	
	0.8	1.7	9	20	110	90	160	90	0.21 ± 0.10
	0.70	1.71			120	105	152	78.3	
	0.2	-0.9	5	40	140	-50	70	160	-0.04 ± 0.09
	0.10	-0.90			130	-43	110	169	

Luminosity Determination in Realistic Scenarios

p_{lab}	x	y	x, y_{cut} around real IP	z_{cut}	tilt_x	tilt_y	div_x	div_y	$\Delta L/L$
(GeV/c)	(mm)				(μrad)				(%)
4.06	0.5	0	3	50	40	−150	80	180	0.08 ± 0.18
	0.44	0.0			48.1	−155	33.9	170	
	−2.0	−0.1	3	20	30	−200	20	110	-0.06 ± 0.18
	−2.06	−0.10			35.4	−204	11.8	91.9	
	−1.0	0	4	30	80	70	20	170	-0.01 ± 0.10
	−1.07	0.01			86.2	67.0	17.5	151	
	0.4	0.8	7	30	−10	−160	0	10	-0.21 ± 0.16
	0.34	0.81			−3.5	−162	6.6	7.5	
	−0.3	1.4	7	30	−110	−10	100	140	-0.03 ± 0.17
	−0.37	1.41			−102	−13.2	97.9	141	

Luminosity Determination in Realistic Scenarios

p_{lab}	x	y	x, y_{cut} around real IP	z_{cut}	tilt_x	tilt_y	div_x	div_y	$\Delta L/L$
(GeV/c)	(mm)				(μrad)				(%)
4.06	-1.6	-0.5	8	20	-130	-190	20	150	0.20 ± 0.12
	-1.66	-0.50			-125	-194	14.3	148	
	1.7	0.6	4	50	-170	120	10	150	-0.30 ± 0.16
	1.63	0.61			-163	118	86.2	176	
	-0.4	1.4	7	20	-170	140	60	150	0.12 ± 0.10
	-0.48	1.41			-164	136	45.4	131	
	-1.1	0.8	5	20	-40	-50	170	150	0.05 ± 0.17
	-1.16	0.80			-34.1	-53.6	157	130	
	-1.2	-0.9	5	10	170	170	20	90	-0.17 ± 0.10
	-1.27	-0.90			179	167	17.5	99.7	

Luminosity Determination in Realistic Scenarios

p_{lab}	x	y	x, y_{cut} around real IP	z_{cut}	tilt_x	tilt_y	div_x	div_y	$\Delta L/L$
(GeV/c)	(mm)				(μrad)				(%)
8.90	-0.6	-1.6	4	40	-140	170	190	10	-0.04 ± 0.11
	-0.68	-1.60			-143	182	159	34.7	
	-0.9	0.7	3	60	190	20	200	110	-0.08 ± 0.11
	-0.98	0.71			203	21.2	184	145	
	-0.2	-1.0	5	60	-40	-190	190	150	-0.03 ± 0.10
	-0.28	-1.0			-33.4	-200	163	101	
	-0.2	1.8	10	30	-180	-80	70	90	0.07 ± 0.56
	-0.28	1.81			-171	-82.4	11.3	10.2	
	-0.5	1.3	6	20	-190	90	80	20	0.41 ± 0.71
	-0.58	1.31			-180	87.1	93.3	15.1	

Luminosity Determination in Realistic Scenarios

p_{lab}	x	y	x, y_{cut} around real IP	z_{cut}	tilt_x	tilt_y	div_x	div_y	$\Delta L/L$
(GeV/c)	(mm)				(μrad)				(%)
8.90	-0.4	0.9	9	10	-30	120	160	60	0.05 ± 0.10
	-0.48	0.91			-143	182	160	75.2	
	-0.8	-1.0	6	10	-190	110	170	80	0.08 ± 0.70
	-0.88	-1.0			-187	115	115	87.3	
	1.0	1.4	9	60	200	80	10	110	0.09 ± 0.11
	0.93	1.41			210	82.8	25.1	52.7	
	-0.8	1.0	8	30	0	150	60	170	0.05 ± 0.11
	-0.88	1.01			11.7	157	120	143	
	1.3	-1.6	3	50	180	-100	60	160	0.06 ± 0.13
	1.23	-1.61			196	-102	12.9	139	